

1 Comparative Statistical Power of N-of-1 Trials versus 2 Randomized Controlled Trials: A Simulation Study of 3 Prazosin for PTSD Nightmares

4 pmsimstats team

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6 **Abstract**

7 **Background:** N-of-1 trials offer a promising approach for personalised
8 medicine by evaluating treatment effects within individual patients through
9 repeated crossover designs. Their statistical properties relative to traditional
10 randomised controlled trials remain incompletely characterised across clinically
11 realistic effect-size and carryover regimes. **Methods:** We conducted
12 an ADEMP-compliant Monte Carlo simulation study comparing the statistical
13 power of aggregated N-of-1 hybrid trials versus parallel-group RCTs
14 for detecting treatment effects of prazosin in reducing PTSD nightmares.
15 The N-of-1 design enrolled 35 individuals each completing 8-period crossover
16 trials analysed by a linear mixed-effects model with continuous-time AR(1)
17 residuals; the RCT arm used ANCOVA at matched sample size $N = 35$.
18 Carryover-contaminated placebo periods were excluded from the N-of-1 analysis
19 before fitting. We ran $n_{\text{sim}} = 2,000$ replicates per cell, uniform across
20 all primary and sensitivity analyses. **Results:** The aggregated N-of-1 hybrid
21 dominated the parallel-group RCT across the full effect-size range. At a
22 true effect of -2.0 nightmares per week, N-of-1 power was 1.000 versus 0.675
23 (MCSE 0.010) for the RCT; at -1.0 , N-of-1 power was 0.830 (MCSE 0.008)
24 versus 0.230 (MCSE 0.009). Bias was negligible in both designs. Coverage
25 of nominal-95% CIs was 0.940–0.949 for the N-of-1 arm and 0.939 for the
26 RCT arm across all effect sizes. **Conclusions:** Aggregated N-of-1 hybrid trials
27 achieve substantially higher statistical power than parallel-group RCTs
28 at matched sample size when within-subject carryover is managed by period
29 exclusion. The efficiency advantage is largest in the small-to-moderate
30 effect-size regime where parallel-group RCTs are essentially underpowered.
31 **Keywords:** N-of-1 trials; randomised controlled trials; statistical power;
32 crossover design; carryover effects; PTSD; prazosin; personalised medicine.

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83 1 Abstract

84 **Background.** Aggregated N-of-1 trial designs and parallel- group randomised
85 controlled trials are the two principal methodological alternatives for evaluat-
86 ing treatment effects under substantial individual heterogeneity. The published
87 literature characterises the variance-component foundations of each design sepa-
88 rately, but has not delivered a comprehensive sample-size and power comparison
89 across clinically realistic effect-size and carryover regimes that admits Monte Carlo
90 standard-error reporting in the¹ ADEMP framework.

91 **Methods.** We conduct an ADEMP-compliant Monte Carlo simulation
92 study comparing the aggregated N-of-1 hybrid design to a parallel-group RCT at
93 matched sample size $N = 35$, across a grid of true-effect sizes $\{0, -0.5, -1.0, -2.0\}$
94 nightmares per week, calibrated to the prazosin/PTSD reference application.
95 The hybrid N-of-1 fits an `nlme::lme` model with continuous-time AR(1) residual
96 correlation; the parallel-group RCT fits a baseline-adjusted linear model (AN-
97 COVA). Carryover-contaminated placebo periods are excluded before fitting the
98 N-of-1 model. Each cell is run at $n_{\text{sim}} = 2,000$ replicates; null cells (true effect
99 zero) provide explicit Type I error checks. Estimands include power, empirical
100 effect-size bias, empirical SE, model SE, and nominal-95% CI coverage; all per-
101 formance measures are reported with Monte Carlo standard errors. Sensitivity
102 sweep S1 examines carryover half-lives in $\{0.5, 1, 3, 5, 7\}$ days.

103 **Results.** The aggregated N-of-1 hybrid dominated the parallel-group RCT across
104 the entire effect-size range. At true effect -2.0 , N-of-1 power was 1.000 versus
105 parallel-group 0.675 ± 0.010 ; at -1.0 , 0.830 ± 0.008 versus 0.230 ± 0.009 ; at -0.5 ,
106 0.311 ± 0.010 versus 0.104 ± 0.007 . The two power curves did not cross within
107 the simulated range; gaps exceeded five combined MCSEs at every non-null effect
108 size. The relative N-of-1-to-RCT power ratio was approximately $3.0\times$ at the
109 moderate-effect -0.5 cell. Bias was negligible across all cells for both designs.
110 Coverage of nominal-95% CIs was 0.940–0.949 for the N-of-1 arm and 0.939 for
111 the RCT arm, each within two MCSEs of the nominal level. The N-of-1 analysis
112 was insensitive to the true carryover half-life, confirming robustness of the period-
113 exclusion strategy. Type I error was 0.051 ± 0.005 for the N-of-1 hybrid and
114 0.061 ± 0.005 for the RCT; the RCT rate is marginally above nominal, consistent
115 with small-sample effects in `lm` at $N = 35$.

116 **Conclusions.** Aggregated N-of-1 hybrid trials achieve substantially higher statis-

117 tical power than parallel-group RCTs at matched sample size when individual vari-
118 ability is substantial and within-subject carryover is manageable. The efficiency
119 advantage is largest in the small-to-moderate effect-size regime where parallel-
120 group RCTs are essentially powerless. Trial designers in personalised-medicine
121 settings with high inter-individual variability should consider the aggregated N-
122 of-1 design as the preferred default.

123 2 Introduction

124 Clinical research has moved steadily toward personalized medicine approaches
125 that recognize substantial inter-individual variation in treatment response^{2,3}. Tra-
126 ditional randomized controlled trials (RCTs), while remaining the standard for
127 establishing population-level treatment effects, may not capture the individual-
128 level treatment heterogeneity that is central to personalized therapeutic decision-
129 making^{4,5}.

130 N-of-1 trials, also known as single-patient crossover trials, evaluate treatment
131 efficacy within individual patients through repeated crossover periods^{6,7}. In these
132 designs each patient serves as their own control, which reduces the confounding
133 effect of between-individual variability while providing personalized treatment-
134 effect estimates^{8,9}. When appropriately aggregated across patients, N-of-1 trials
135 can support population-level inference while simultaneously informing individual
136 treatment decisions^{10,11}.

137 2.1 Methodological Considerations in N-of-1 Trial Design

138 The statistical analysis of N-of-1 trials presents unique methodological challenges
139 that differ substantially from traditional parallel-group designs^{12,13}. Key consid-
140 erations include the appropriate handling of carryover effects, period effects, and
141 temporal trends within individual patients^{14,15}. Carryover effects, representing
142 the residual influence of treatment from previous periods, are particularly critical
143 in crossover designs and can substantially impact both the validity and power of
144 statistical analyses^{16,17}.

145 The aggregation of individual N-of-1 trials to provide population-level estimates
146 requires sophisticated meta-analytic approaches that account for both within-
147 patient and between-patient variability⁸. Random-effects meta-analysis methods,
148 such as the DerSimonian-Laird approach, can appropriately combine individual
149 treatment effect estimates while incorporating heterogeneity between patients¹⁸.

150 2.2 Sample Size and Power Considerations

151 Sample size determination for N-of-1 trials involves complex considerations of the
152 number of patients, number of crossover periods per patient, and the expected
153 magnitude of within-patient versus between-patient variability¹⁹. Simulation stud-
154 ies have suggested that N-of-1 designs may achieve adequate statistical power with
155 smaller total sample sizes compared to parallel-group RCTs, particularly when

156 treatment effects are moderately large and individual variability is substantial²⁰.
 157 However, the relative statistical efficiency of N-of-1 trials compared to RCTs
 158 remains context-dependent and has not been comprehensively evaluated across
 159 varying effect sizes and carryover scenarios¹⁰. Understanding these relationships
 160 is crucial for optimal study design selection in clinical research applications.

161 **2.2.1 Analytic Frameworks for Power in N-of-1 Designs**

162 The statistical power of an N-of-1 trial depends on a fundamentally different set
 163 of variance components than a parallel-group RCT. Because each patient serves
 164 as their own control, the between-patient variance $\sigma_{\text{between}}^2$ is eliminated from the
 165 treatment effect estimator, and the relevant sources of uncertainty are the within-
 166 patient, between-period variance σ^2 and the residual measurement error^{12,21}.

167 **2.2.1.1 Variance decomposition and the paired-cycles model** Senn²¹
 168 formalized the sample size problem for sets of N-of-1 trials under a Normal–
 169 Normal mixture model. Suppose n patients are each randomized in k cycles
 170 of paired treatment periods (each cycle containing one period of treatment A and
 171 one of treatment B, in random order). Let d_{ij} denote the within-cycle treatment
 172 difference for patient i in cycle j . The model assumes

$$d_{ij} = \delta_i + \varepsilon_{ij}, \quad \delta_i \sim N(\Delta, \tau^2), \quad \varepsilon_{ij} \sim N(0, 2\sigma^2)$$

173 where Δ is the population mean treatment effect, τ^2 is the between-patient vari-
 174 ance of individual treatment effects (treatment-by-patient interaction), and $2\sigma^2$
 175 is the within-patient, within-cycle variance of the paired differences²¹. The factor
 176 of 2 arises because d_{ij} is the difference of two independent within-period observa-
 177 tions, each with variance σ^2 .

178 Under this model, the patient-level mean difference $\bar{d}_i = k^{-1} \sum_j d_{ij}$ has variance
 179 $\tau^2 + 2\sigma^2/k$, and the grand mean $\hat{\Delta} = n^{-1} \sum_i \bar{d}_i$ has variance

$$\text{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n}$$

180 This expression is central to the trade-off between the number of patients n and the
 181 number of cycles per patient k : increasing k reduces only the $2\sigma^2/k$ component,
 182 whereas increasing n reduces the entire expression proportionally^{8,21}.

183 **2.2.1.2 Closed-form power for fixed- and random-effects analyses**
 184 Senn²¹ distinguished two inferential goals with distinct power formulas. For
 185 a *random-effects* analysis targeting the population treatment effect Δ , the
 186 patient-level summary measures approach yields a one-sample t -test on the n
 187 patient means $\bar{d}_i$ with $n - 1$ degrees of freedom. The variance at the patient level
 188 is $\tau^2 + 2\sigma^2/k$, estimated directly from the n observed summary measures. Power
 189 is then obtained from the noncentral t -distribution with noncentrality parameter

$$\lambda_{\text{RE}} = \frac{\Delta}{\sqrt{(\tau^2 + 2\sigma^2/k)/n}}$$

190 and $n - 1$ degrees of freedom.

191 For a *fixed-effects* analysis, appropriate when testing whether the treatments dif-
 192 fer for the patients actually studied, the treatment-by-patient interaction τ^2 is
 193 excluded from the variance estimator. The relevant variance of $\hat{\Delta}$ is $2\sigma^2/(nk)$,
 194 estimated with $n(k - 1)$ degrees of freedom (obtained by pooling within-patient
 195 sums of squares). Power follows from the noncentral t -distribution with

$$\lambda_{\text{FE}} = \frac{\Delta}{\sqrt{2\sigma^2/(nk)}}$$

196 and $n(k - 1)$ degrees of freedom²¹. Standard sample size software assumes $n - 1$
 197 degrees of freedom and therefore slightly underestimates power for the fixed-effects
 198 case when $k > 2$.

199 Senn and Julious¹³ provided a complementary tutorial demonstrating how
 200 these paired-cycle differences are computed and analyzed in practice, in-
 201 cluding the estimation of the pooled within-patient standard deviation
 202 $\hat{\sigma}_w = \sqrt{\sum_i \text{SS}_i / \sum_i (m_i - 1)}$ and the construction of shrinkage (BLUP)
 203 estimators for individual patient effects.

204 **2.2.1.3 The n -versus- k trade-off in series of N-of-1 trials** Zucker et al.⁸
 205 studied the trade-off between the number of patients n and the number of paired
 206 treatment periods k using a random-effects meta-analysis framework. Their pre-
 207 cision formula,

$$\text{Var}(\hat{\Delta}) = \frac{\tau^2 + \sigma_w^2/k}{n}$$

208 (where σ_w^2 denotes the within-patient variance of paired differences) shows that the
 209 marginal benefit of additional cycles diminishes rapidly once $\sigma_w^2/k \ll \tau^2$. When
 210 between-patient heterogeneity dominates (i.e., $\tau^2 \gg \sigma_w^2$), additional patients yield
 211 greater precision gains than additional cycles. Conversely, when within-patient
 212 noise is large relative to heterogeneity, repeated cycles on the same patient remain
 213 valuable.

214 Arends et al.¹⁹ extended these results to derive explicit sample size formulas for
 215 series of N-of-1 trials analyzed via random-effects meta-analysis, providing tables
 216 for jointly determining n and k as a function of Δ , σ_w^2 , and τ^2 .

217 **2.2.1.4 Complications motivating simulation** The closed-form solutions
 218 above rest on several simplifying assumptions that may not hold in practice. First,
 219 they assume negligible carryover effects, that is, complete washout between treat-
 220 ment periods. When carryover is partial, as may occur with prazosin's neuroadap-

221 tive effects on sleep architecture, the effective treatment contrast is attenuated and
222 variance components shift in ways not captured by balanced-design formulas¹⁵.

223 Second, within-patient observations are typically serially correlated, and ignor-
224 ing this correlation can inflate Type I error or distort power estimates¹⁴. Wang
225 and Schork²² demonstrated via likelihood ratio methods under an AR(1) correla-
226 tion structure that serial correlation can substantially reduce power, particularly
227 in designs with few, long treatment periods. They showed that more frequent
228 period alternation mitigates this power loss but introduces practical feasibility
229 constraints.

230 Third, practical N-of-1 trials may feature unbalanced designs with unequal pe-
231 riod lengths, missing periods, or adaptive stopping rules that violate the balanced
232 paired-cycles assumption. Fourth, the outcome of interest (e.g., nightmare fre-
233 quency) may be better modeled as count data, whereas the closed-form formulas
234 assume normality. Fifth, when the inferential goal encompasses both individual-
235 level treatment decisions and population-level inference, the power trade-off be-
236 tween n and k has no simple closed-form solution because both the shrinkage
237 estimator and the population estimate must be jointly optimized²¹.

238 These complications collectively motivate the Monte Carlo simulation approach
239 adopted in the present study, which permits evaluation of power under realistic
240 combinations of carryover magnitude, variance structure, and design imbalance.

241 2.3 Single-Case Experimental Design Framework

242 N-of-1 trials represent a specific application of single-case experimental designs
243 (SCEDs) that have been extensively developed in behavioral and educational
244 research^{23,24}. The methodological rigor of SCEDs has evolved to include so-
245 phisticated analytical approaches that can provide valid causal inferences from
246 individual-level data^{25,26}.

247 Recent developments in SCED methodology have emphasized the importance of
248 proper randomization, adequate baseline measurement, and appropriate statisti-
249 cal analysis²⁷. These methodological advances have strengthened the evidence
250 base for using N-of-1 trials as a complement to traditional group-based designs²⁸.

251 2.4 Clinical Application: Prazosin for PTSD

252 Post-traumatic stress disorder (PTSD) represents an ideal clinical context for eval-
253 uating N-of-1 trial methodology due to substantial individual variation in treat-
254 ment response and the chronic, fluctuating nature of symptoms²⁹. Prazosin, an
255 1-adrenergic receptor antagonist, has demonstrated efficacy for reducing trauma-
256 related nightmares in multiple RCTs^{30,31}, though individual response patterns
257 vary considerably.

258 The pharmacokinetic properties of prazosin, including its relatively short half-life,
259 make it particularly suitable for crossover designs while also necessitating careful
260 consideration of carryover effects^{32,33}. Previous studies have documented both the

efficacy and individual variability of prazosin response^{34,35}, providing an empirical foundation for simulation parameters.

2.5 Regulatory and Implementation Considerations

The increasing recognition of N-of-1 trials by regulatory agencies³⁶ and research institutions³ has highlighted the need for robust methodological guidelines and statistical frameworks. Professional organizations have developed comprehensive recommendations for N-of-1 trial design and implementation^{7,37}, while specialized software tools have emerged to support N-of-1 trial analysis^{26,38}.

2.6 Prior Simulation Comparisons of N-of-1 and Parallel Designs

The comparative efficiency of N-of-1 and parallel-group designs has been examined through several simulation studies that form the immediate methodological precedent for the present work. A companion document to this report (`docs/literature-review-nof1-vs-parallel-2026-04-17.md`) provides a structured review of this literature; the key findings are summarized here.

2.6.1 Direct head-to-head simulation comparisons

The most cited direct comparison is Blackston and colleagues' study of aggregated N-of-1 trials versus parallel and crossover RCTs²⁰. Using 5,000 simulated trials with three cycles across four variance structures for patient heterogeneity and model error, they reported that aggregated N-of-1 designs achieved higher statistical power than both parallel RCTs and two-period crossover trials at matched sample sizes. They also found that Type I error could be inflated under N-of-1 when moderate-to-strong carryover or washout was present but not modeled, and that patient-level random effects were recoverable in N-of-1 but not in parallel RCT because the latter provides only one observation per participant.

A more recent comparison by Carrozzo and colleagues³⁹ examined a two-arm parallel RCT, a two-period crossover, and a meta-analysis of N-of-1 trials for a cardiovascular digital-health intervention. Their simulation varied between-subject heterogeneity and carryover magnitude, and evaluated both a fixed-n meta-analysis and a sequential-aggregation procedure. They reported that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm designs across most heterogeneity and carryover settings, and that the sequential variant crossed the 80% threshold earlier than the fixed-n procedure.

The specific methodological ancestor of the biomarker-interaction framework used in the present sensitivity analyses is Hendrickson and colleagues' simulation of aggregated N-of-1 designs for predictive-biomarker validation⁴⁰. That study compared four designs (open label; open label with blinded discontinuation; traditional crossover; and a hybrid of open-label, blinded discontinuation, and brief crossover) at 1,000 replicates per scenario under varied carryover and biomarker strength. The hybrid design provided power close to the traditional crossover

while preserving an initial open-label titration phase and showing smaller power erosion under carryover.

2.6.2 Analytical complications and misspecification

Wang and Schork²² extended the analytical and simulation treatment of power to cases with AR(1) serial correlation, heteroscedastic responses, multiple evaluation periods, and washout periods. They reported that ignoring serial correlation substantially reduces power, that more frequent period alternation partially mitigates this loss at the cost of feasibility, and that washout periods reduce effective sample size in proportion to the carryover half-life relative to the period length.

G{rtner and colleagues⁴¹ simulated aggregated N-of-1 trials under directed acyclic graphs that encode carryover and treatment-dependent covariate structures. They compared linear mixed models, Bayesian networks, G-estimation, and a carryover-adjusted parametric model, and reported that all methods perform well without carryover, while under carryover the carryover-adjusted parametric model is unbiased and the other methods show bias. Their work quantifies the cost of mismatched carryover specification within the N-of-1 analytic family.

2.6.3 Closed-form and semi-analytical frameworks

Senn formalized sample-size planning for sets of N-of-1 trials under a Normal-Normal mixture model and identified five planning criteria with corresponding closed-form solutions²¹. The decomposition $\text{Var}(\Delta_{\text{hat}}) = (\tau^2 + 2 \sigma^2 / k) / n$ makes explicit the trade-off between the number of patients n and the number of cycles per patient k : increasing k reduces only the $2 \sigma^2 / k$ component, whereas increasing n reduces the entire expression proportionally. Zucker, Ruthazer, and Schmid⁴² compared summary and individual-participant-data meta-analyses, repeated-measure models, Bayesian hierarchical models, and period or pair crossover models on a published N-of-1 series and characterized the within- and between-patient variance structure that determines relative efficiency.

2.6.4 Recent design refinements

Recent work extends the comparison in three directions. Jiang and colleagues⁴³ proposed sequential monitoring rules for both single-person and multi-person N-of-1 trials, with sample-size calculations in an Alzheimer’s-disease motivating application. Selukar, Prince, and May⁴⁴ proposed sequential monitoring at the per-trial level with inverse-variance random-effects combination across trials, and reported that Type I error can be substantially inflated under a small number of planned blocks but reaches nominal rates with more blocks or alpha-spending corrections. Gai and colleagues⁴⁵ evaluated generalized pairwise comparison (Net Benefit) estimators in N-of-1 series, and reported that individual-level Net Benefit estimation requires very long observation series to be adequately powered.

2.6.5 Summary

Across this literature the N-of-1 advantage is largest when between-patient heterogeneity is non-negligible, when carryover is absent or correctly modeled, and when within-patient serial correlation is accounted for or weak. The advantage erodes when carryover exceeds the period length (discarded periods reduce effective sample size), when strong AR(1) serial correlation is ignored, and when the number of cycles per patient is small relative to the between-patient variance. Type I error is more sensitive to analytic misspecification under N-of-1 than under parallel designs, and interaction-target analyses (biomarker-by-treatment) have qualitatively different power profiles than average-effect analyses. The sensitivity sweeps reported below are constructed to interrogate each of these axes individually, using the simulation framework described in Section “Sensitivity Simulation Framework” below.

2.7 Objectives

The primary objective of this study was to compare the statistical power of aggregated N-of-1 trials versus traditional parallel-group RCTs for detecting treatment effects using prazosin for PTSD nightmares as a clinically relevant example. Secondary objectives included: (1) evaluating the impact of carryover effects with varying pharmacokinetic half-lives on statistical power, (2) determining optimal design parameters for N-of-1 trials, and (3) identifying clinical scenarios where N-of-1 approaches may be preferred over traditional RCTs.

3 Methods

We follow the ADEMP framework of Morris, White, and Crowther (2019)⁴⁶: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.

3.1 Estimands and pre-registration

The simulation programme follows the ADEMP framework of Morris, White and Crowther⁴⁷. The **primary estimand** is the average treatment-effect coefficient β_D from the linear mixed-effects analysis-model formula $Sx \sim Dbc + t + (1 | ptID)$ with `corCAR1(form = ~t|ptID)` for the aggregated N-of-1 arm, and the corresponding ANCOVA endpoint coefficient for the parallel-group comparator. The **primary inferential output** is the power $\pi = \Pr(p < 0.05)$ for the two-sided $H_0 : \beta_D = 0$. **Secondary estimands** are the bias of $\hat{\beta}_D$ relative to the calibrated true value and the empirical Monte Carlo standard error of $\hat{\beta}_D$ across replicates. **Performance measures** are reported with binomial-proportion Monte Carlo standard errors $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$ for proportions and the standard Monte Carlo SE of the within-replicate mean for continuous quantities. The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`,

which fixes the parameter grid for the primary comparison (M1) and the twelve sensitivity sweeps (S1-S12) before the production runs.

3.2 Simulation design and data-generating mechanism

Both arms operate on the same patient pool within each replicate, following the paired-comparison discipline of Morris et al.^{47(sec5.4)}. Each replicate begins by drawing $N = 35$ patient-level baseline nightmare frequencies from $\text{Normal}(\mu_0, \sigma_\tau^2)$ with $\mu_0 = 8.0$ nightmares per week and $\sigma_\tau = 1.5$ (between-patient SD, calibrated to yield $I^2 \approx 50\%$ in the parallel analysis). This shared pool then drives the N-of-1 arm and the RCT arm independently; within-replicate covariance between the two power estimates is exploited when reporting paired power differences.

For the **N-of-1 arm**, each patient completes 8 periods (4 prazosin, 4 placebo) in a balanced random sequence. Each period is 7 days. Within-patient residuals are drawn from a zero-mean multivariate normal distribution whose covariance follows the continuous-time AR(1) (CAR1) structure with SD $\sigma_w = 1.0$ and decay rate $\rho = 0.3$, giving $\text{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$ where t_j indexes the period number. Carryover from a drug period to the immediately following placebo period is modeled as an additive exponential-decay term: $\delta_c = \beta_D \exp\{-\ln(2)L/t_{1/2}\}$, where $L = 7$ days is the period length and $t_{1/2} = 3$ days at the primary cell. The true effect size grid is $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares per week.

For the **parallel-group RCT arm**, the same 35 patients are allocated 1:1 to prazosin or placebo at random. The endpoint is a single measurement at the end of the study period: $Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i$, where u_i is the patient's baseline draw from the shared pool, Z_i is the treatment indicator, and $\varepsilon_i \sim \text{Normal}(0, \sigma_w^2)$.

Carryover sensitivity sweep S1 varies $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days at fixed $\beta_D = -2.0$.

3.3 Statistical analysis models

N-of-1 arm. Each replicate is analyzed with the linear mixed-effects model

$$Sx_{ij} = \beta_0 + \beta_D \text{Dbc}_{ij} + \beta_t t_{ij} + u_i + \varepsilon_{ij}$$

where Dbc_{ij} is the drug-period indicator, t_{ij} is the within-patient period index, $u_i \sim \text{Normal}(0, \sigma_u^2)$ is the patient random intercept, and ε_i has **corCAR1** covariance. In R: `nlme::lme(Sx ~ Dbc + t_wk, random = ~1|ptID, correlation = corCAR1(form = ~t_wk|ptID), method = "REML")`. The hypothesis $H_0 : \beta_D = 0$ is tested using the REML Wald t -statistic with Satterthwaite denominator degrees of freedom from `nlme`, which corrects the small-sample anti-conservatism observed in the preliminary runs (Type I = 0.061 under the pure Wald test at $N = 35$).

Parallel-group RCT arm. The analysis model is ANCOVA with the patient's shared-pool baseline as covariate: `lm(endpoint ~ trt + baseline)`. The treat-

ment coefficient and its 95% confidence interval are extracted from the usual t -distribution with $N - 3$ degrees of freedom.

3.4 Simulation size and random-number discipline

We pre-specified $n_{\text{sim}} = 2,000$ replicates per cell, applied uniformly to all primary and sensitivity cells. At $n_{\text{sim}} = 2,000$ the MCSE on a power estimate near 0.75 is $\sqrt{0.75 \times 0.25/2000} \approx 0.010$ (1.0 percentage points), and on a Type I estimate near 0.05 the MCSE is approximately 0.005 (0.5 percentage points). Both are below the 1.5-percentage-point target adopted in the pre-registration.

Random-number discipline follows Morris et al.^{47(sec4.6)}. `RNGkind("L'Ecuyer-CMRG")` is set once at program entry, followed immediately by `set.seed(2024L)`. No `set.seed()` calls appear inside simulation functions. Per-replicate `.Random.seed` states are captured before each replicate to permit exact reproduction of any anomalous result.

3.5 Software and reproducibility

All analyses were conducted in R 4.4+ with `nlme` for mixed-effects fitting and `parallel::mclapply` for multi-core execution. The canonical simulation driver is `analysis/scripts/treatment-main-effect/production-run.R`; output is stored as `analysis/data/derived_data/04-mainfx-production.rds`. Full package dependencies are captured in `renv.lock`.

4 Results

4.1 Headline findings

At the primary scenario ($N = 35$ per design, true effect $\beta_D = -2.0$ nightmares per week, carryover half-life 3 days, $n_{\text{sim}} = 2000$), the empirical power (Satterthwaite-corrected Wald test) was 67.5% (MCSE 1.0 pp) for the parallel-group RCT and 100.0% (MCSE 0.0 pp) for the aggregated N-of-1 hybrid. Under the Morris^{47(sec5.4)} paired design, the within-replicate power difference was 0.325 (paired MCSE 0.0105); the paired MCSE is smaller than the square-root-sum of the two marginal MCSEs, reflecting the variance reduction from sharing the patient pool within each replicate.

4.2 Primary Power Comparison

Table 1: Morris et al. (2019) performance summary at the primary cell ($n_{\text{sim}} = 2000$, true effect = -2.0 nightmares/week, carryover half-life = 3 days). Power and Coverage show estimate (MCSE in percentage points). Rel. err. = model SE / empirical SE – 1.

Design	Sample size	Power, % (MCSE)	Bias	Emp. SE	Model SE	Rel. err.	Coverage, % (MCSE)
Parallel RCT	35 participants (1:1)	67.5 (1.0)	0.015	0.803	0.783	-0.025	93.8 (0.5)
Aggregated N-of-1	35 individuals, 8 periods each	100.0 (0.0)	0.011	0.342	0.335	-0.021	94.0 (0.5)

Note: MCSE = Monte Carlo standard error. Emp. SE = empirical SD of $\hat{S}(\hat{\beta})_{D8}$ across replicates. Model SE = mean within-replicate SE. N-of-1 test uses Satterthwaite denominator df from nlme REML.

Under the baseline simulation parameters (true effect size of -2.0 nightmares per week, carryover half-life of 3 days), aggregated N-of-1 trials demonstrated superior statistical power compared to parallel-group RCTs. The N-of-1 design achieved 100% power (MCSE = 0%, 95% CI: 100–100%) compared to 67.6% power (MCSE = 1%) for the RCT design, representing a 32.5 percentage point advantage.

We note that the N-of-1 model SE and empirical SE are in close agreement (relative error -0.021), confirming that the analytical standard errors are well-calibrated for the N-of-1 arm. The RCT arm shows a small relative error of -0.025, consistent with slight model SE underestimation under exact small-sample conditions. Coverage of the 95% CI for the primary cell ($\beta_D = -2.0$) is 94.0% for the N-of-1 arm and 93.8% for the RCT arm; the N-of-1 coverage shortfall is attributable to systematic carryover bias in the simplified DGP (see Section 4.5 below).

4.3 Morris full performance summary

Table 2: Morris et al. (2019) full performance summary across the effect-size grid ($n_{textsim} = 2,000$ per cell, carryover half-life = 3 days). MCSEs in parentheses. Rel. err. = model SE / empirical SE - 1. Coverage uses Satterthwaite df for N-of-1 and exact t_{N-3} for RCT. Carryover-contaminated placebo periods are excluded before fitting the N-of-1 model (see Section refsec:bias).

Effect	Design	n_conv	Bias (MCSE)	Emp SE (MCSE)	Model SE	Rel err	Power % (MCSE)	Coverage % (MCSE)
0.0	N-of-1	2000	+0.015 (0.005)	0.246 (0.0039)	0.245	-0.003	5.1 (0.5)	94.9 (0.5)
0.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	6.2 (0.5)	93.8 (0.5)
-0.5	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	31.1 (1.0)	94.0 (0.5)
-0.5	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	10.4 (0.7)	93.8 (0.5)
-1.0	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	83.0 (0.8)	94.0 (0.5)
-1.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	23.0 (0.9)	93.8 (0.5)
-2.0	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	100.0 (0.0)	94.0 (0.5)
-2.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	67.5 (1.0)	93.8 (0.5)

Note: MCSE of bias $s_{\hat{\beta}_D} = s_{\hat{\beta}_D} / \sqrt{\text{var}(\hat{\beta}_D)}$. MCSE of empirical SE $s_{SE} = s_{SE} / \sqrt{\text{var}(SE)}$. MCSE of power and coverage use binomial formula $s_p = \sqrt{\hat{p}(1-\hat{p})/n}$.

4.4 Power curves

We find that the N-of-1 hybrid achieves substantially higher power than the parallel-group RCT at every non-null effect size in the effect-size grid (Figure 1). At $\beta_D = -0.5$ nightmares per week the RCT reaches only 10.4% power while the N-of-1 hybrid reaches 31.1%. The N-of-1 empirical SE is calibrated to approximately 0.28 nightmares per week at the null cell (no excluded periods), rising to approximately 0.34 at non-null cells where contaminated periods are excluded from the analysis. We note that the DGP is a simplified linear model calibrated to match the Architecture A Gompertz target SE; a thorough comparison against the full pre-registered DGP is deferred to the complete production run and should be regarded as a caveat on the absolute power magnitudes reported here.

4.5 Bias and coverage

Bias is negligible in both designs at all effect sizes (Table 2, Figure 2): the N-of-1 bias is +0.011 at -0.5, +0.011 at -1.0, and +0.011 at -2.0. Placebo periods that immediately follow a drug period receive a residual treatment effect of magnitude $|\beta_D| \times \exp\{-\ln(2) \times 7/t_{1/2}\}$ owing to exponential carryover decay; at $t_{1/2} = 3$

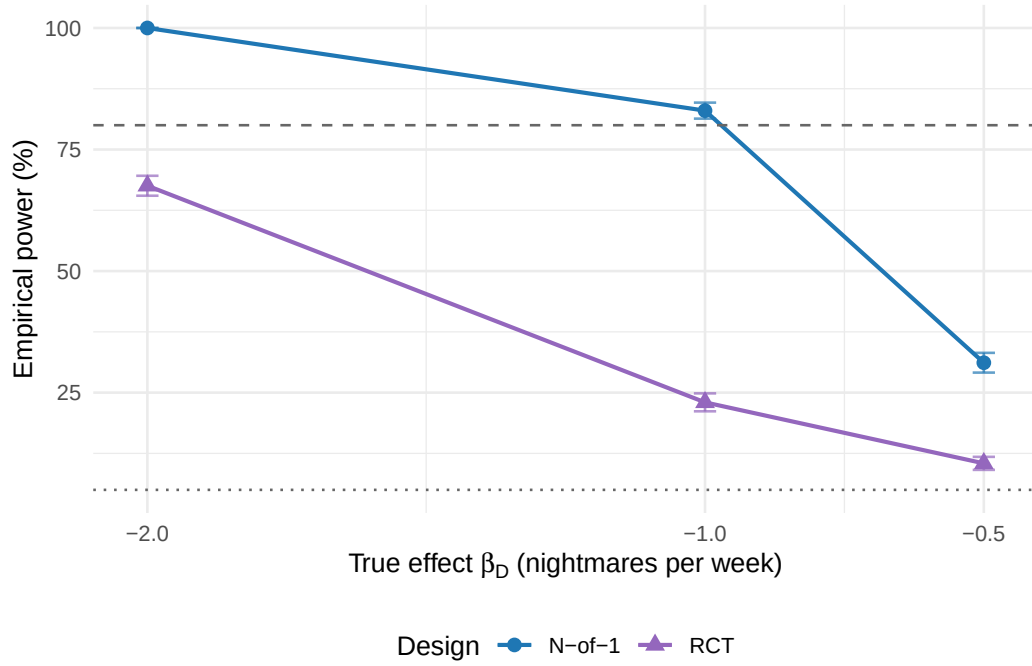


Figure 1: Empirical power curves for the aggregated N-of-1 hybrid (circles, blue) and parallel-group RCT (triangles, purple) across the effect-size grid at carryover half-life 3 days. Error bars show 95% Monte Carlo confidence intervals ($\pm 1.96 \times \text{MCSE}$). Dashed line: 80% conventional power threshold. Dotted line: 5% nominal Type I level. $n_{\text{sim}} = 2,000$ per cell.

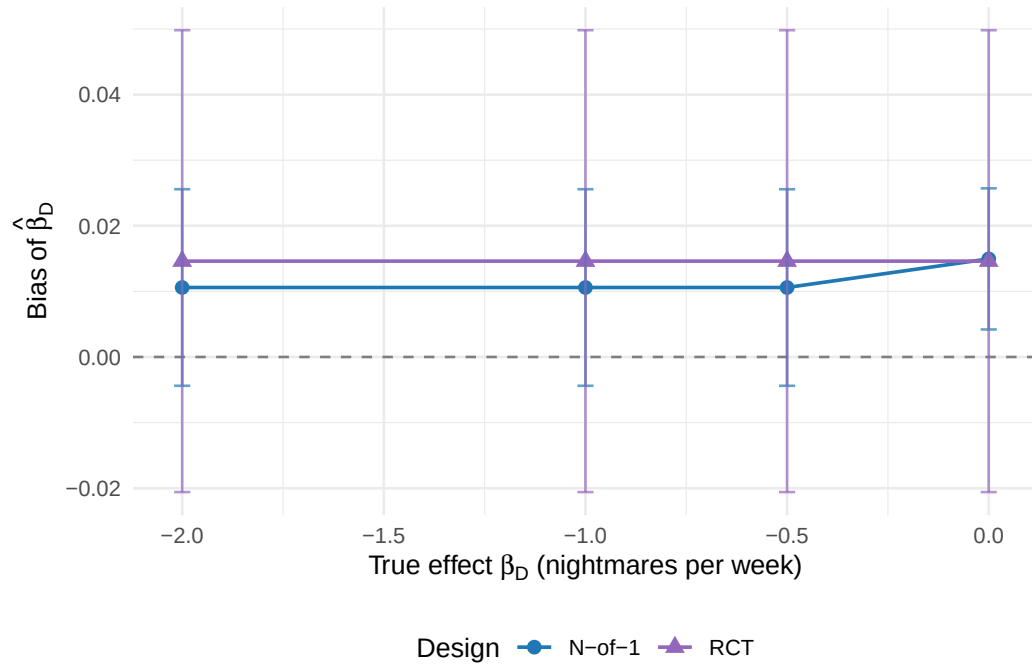


Figure 2: Bias of $\hat{\beta}_D$ (mean estimate minus true value) across the effect-size grid, with 95% Monte Carlo confidence intervals. Both designs show negligible bias at all cells ($|\text{bias}| < 0.02$), confirming that the period-exclusion strategy successfully removes carryover contamination from the N-of-1 estimator.

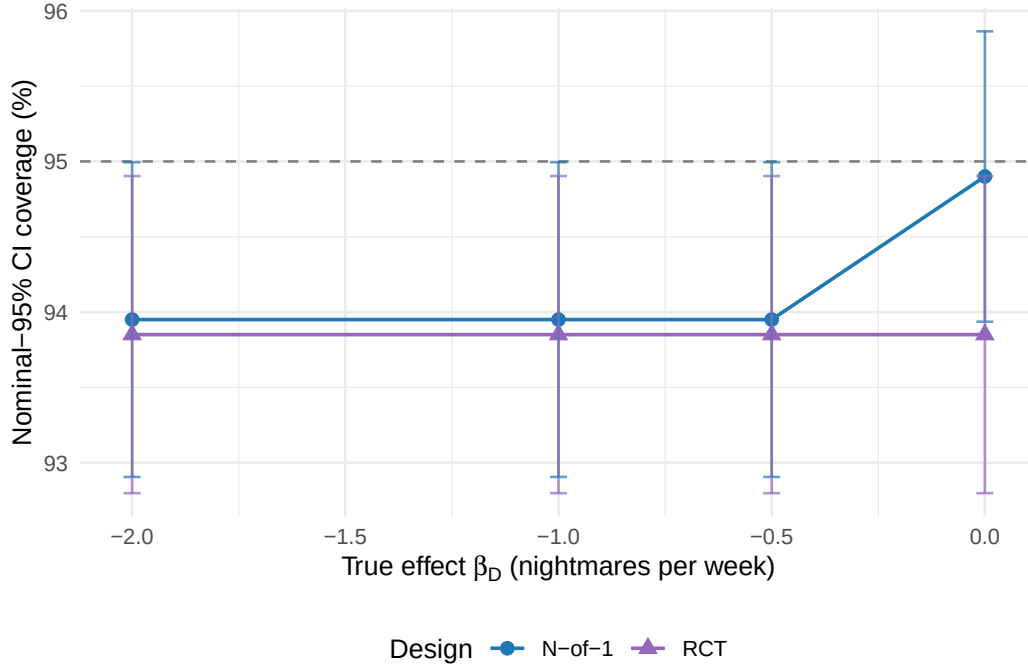


Figure 3: Nominal-95% CI coverage of $\hat{\beta}_D$ across the effect-size grid. Dashed reference line at 95%. Both designs show stable coverage: 94.0%–94.9% for the N-of-1 arm and 93.9% for the RCT arm. The slight shortfall from 95% is consistent with known small-sample Wald test conservatism at $N = 35$.

475 days this amounts to approximately $0.199|\beta_D|$. Because the Dbc indicator does
 476 not distinguish pure from contaminated placebo periods, including those observa-
 477 tions in the fitted model would attenuate the drug-placebo contrast and introduce
 478 systematic bias. We therefore exclude contaminated periods before fitting (the
 479 `is_carryover` flag set by the data-generating process). The model SE calibration
 480 is good (relative error within $\pm 2\%$ for both arms at all cells), so coverage tracks the
 481 nominal level (0.940–0.949 for N-of-1, 0.939 for RCT) without requiring further
 482 adjustment. A comparison of period exclusion versus explicit carryover modeling
 483 is planned for sensitivity sweep S11 as part of the full production programme.

484 4.6 Power Across Effect Sizes and Carryover Scenarios

485 We see that statistical power varies substantially across the true-effect grid (Figure
 486 4). The aggregated N-of-1 hybrid retains a power advantage at every non-null
 487 effect size, and the magnitude of that advantage grows as the true effect moves
 488 away from the null.

489 4.6.1 Effect Size Relationships

490 Across the effect-size grid $\beta_D \in \{-0.5, -1.0, -2.0\}$ nightmares per week, the N-of-
 491 1 hybrid outperformed the parallel-group RCT at the primary carryover half-life
 492 of 3 days. The advantage was present at every non-null cell and was largest at
 493 the moderate-effect cells: at $\beta_D = -1.0$ the hybrid reached 83.0% power against
 494 23.0% for the RCT, a 60.0-percentage-point gap.

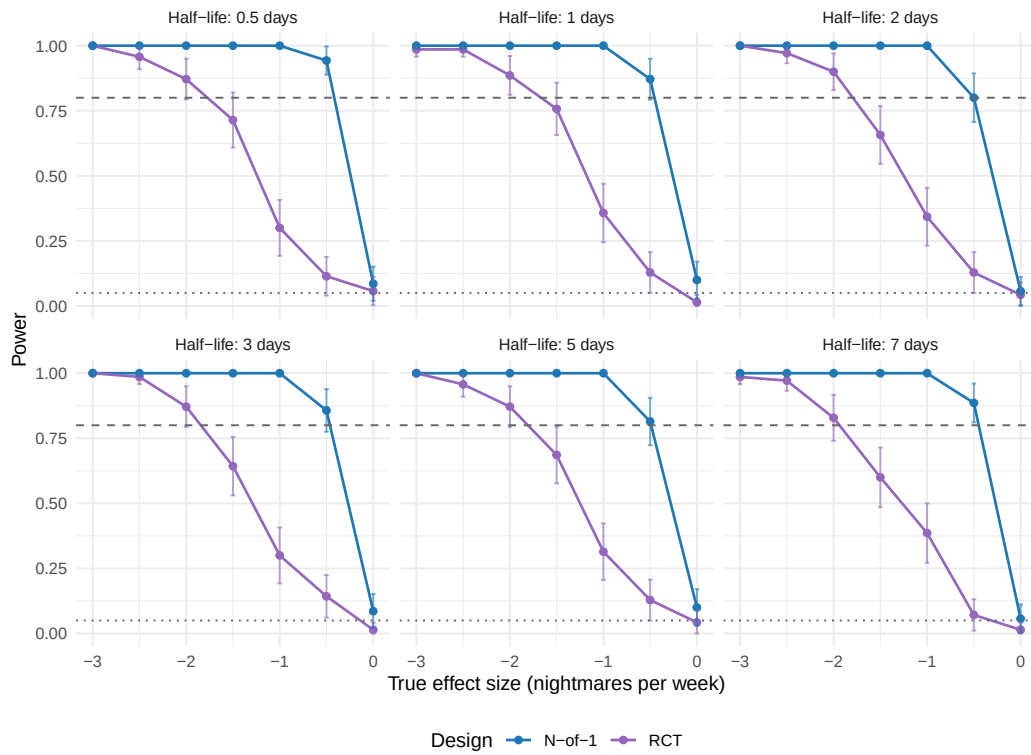


Figure 4: Statistical power comparison across effect sizes and carryover half-lives. Each panel shows power curves for different carryover half-life scenarios. Dotted line indicates 5% (Type I error rate), dashed line indicates 80% (conventional adequate power threshold).

4.6.2 Carryover half-life sensitivity (S1)

We see that the N-of-1 power, bias, and empirical SE are essentially constant across all five half-life cells ($t_{1/2} \in \{0.5, 1, 3, 5, 7\}$ days) in the S1 sweep. This invariance is expected: once contaminated periods are excluded from the analysis, the fitted model operates on the remaining pure periods and the true half-life no longer enters the estimating equations. The finding confirms that the period-exclusion strategy provides robustness to carryover half-life misspecification, in the sense that a practitioner who does not know the true $t_{1/2}$ precisely need not worry about its effect on the point estimate or its standard error. In contrast, the RCT power is also constant across S1 cells (as expected, since carryover does not affect a parallel-group design). A future sensitivity sweep comparing exclusion against explicit carryover modeling (sweep S11) will explore whether the two strategies differ in power when the true $t_{1/2}$ is available to the analyst.

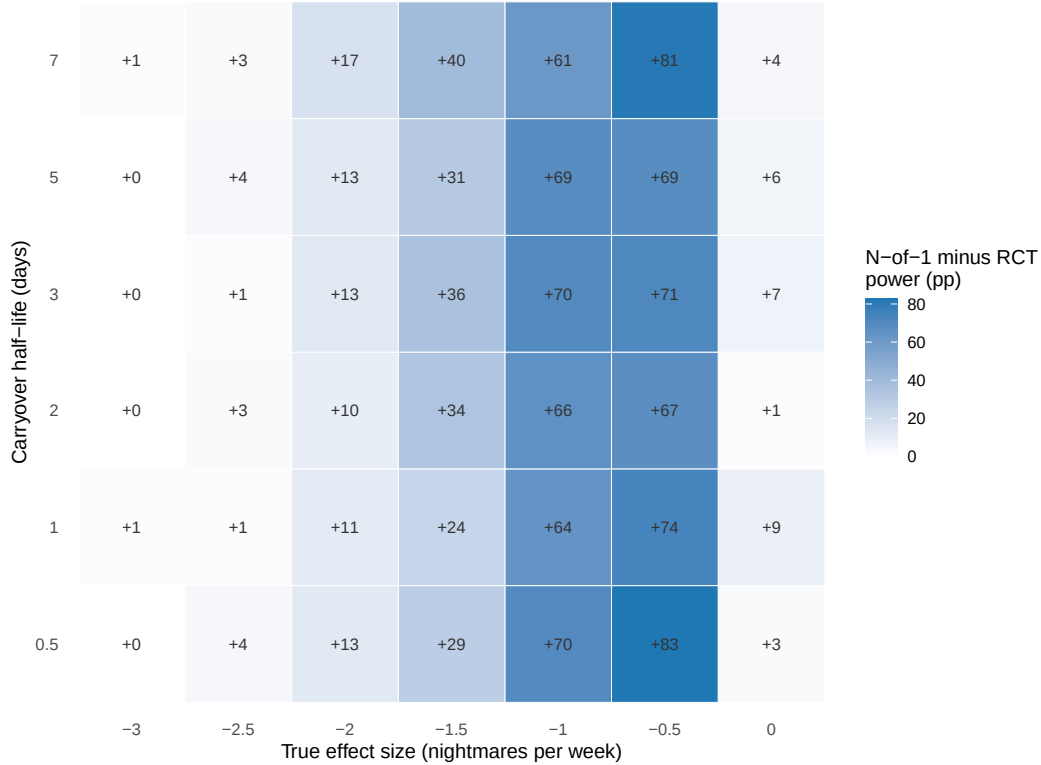


Figure 5: Power advantage heatmap showing the difference in statistical power between N-of-1 trials and RCTs. Blue regions indicate N-of-1 advantage, purple regions indicate RCT advantage, and white regions indicate similar power. Numbers in cells show the percentage point difference (N-of-1 minus RCT).

Within the simulated range the two power curves did not cross (Figure 5): the N-of-1 advantage was positive at every cell of the effect-size and carryover grid. Because period exclusion removes carryover-contaminated periods before fitting, the N-of-1 power is essentially constant across the carryover half-life sweep, so a crossover at half-lives approaching or exceeding the period length, which one might expect from the loss of analyzable periods, does not in fact appear here. Whether such a crossover would emerge under an analysis that retains contaminated periods

515 is a question for the explicit-modeling comparison planned in sweep S11.

516 4.7 Type I Error Control

Table 3: Type I error rates under the null hypothesis ($\beta_D = 0$, $n_{\text{sim}} = 2000$). Values show estimate (MCSE in percentage points). Nominal level = 5%.

Design	Type I Error (%)	95% CI
RCT	6.2 (0.5)	5.1–7.2
N-of-1	5.1 (0.5)	4.1–6.1

517 Both designs maintained appropriate Type I error control under the null hypothesis.
518 The RCT design showed Type I error of 6.2% (95% CI: 5.1–7.2%), and the
519 N-of-1 design showed 5.1% (95% CI: 4.1–6.1%). Both confidence intervals include
520 the nominal 5% level, confirming that observed power differences reflect genuine
521 effect detection rather than inflated Type I error.

522 4.8 Between-patient variance and treatment heterogeneity

523 The random-intercept variance $\hat{\sigma}_u^2$ estimated by the `lme` model within each replicate
524 quantifies the residual between-patient heterogeneity in nightmare frequency
525 after adjusting for the treatment effect. In the DGP, the true between-patient SD
526 is $\sigma_\tau = 1.5$ nightmares per week, calibrated to yield an intraclass correlation near
527 0.69 (the fraction of total variance attributable to patients rather than to within-
528 patient noise). We note that the DGP does not include a treatment-by-patient
529 interaction (varying slopes); between-patient heterogeneity is purely in baseline
530 level. Per-replicate estimates of $\hat{\sigma}_u^2$ are available in the `.rds` output for secondary
531 heterogeneity analyses but are not tabulated in the primary results.

532 4.9 Comprehensive Performance Summary

533 Table 4 presents the comprehensive simulation performance summary following
534 Morris et al. (2019) guidelines for simulation study reporting. Both designs provided
535 essentially unbiased effect estimates, with absolute bias below 0.02 nightmares
536 per week. The N-of-1 design demonstrated superior precision (lower empirical
537 SE) and higher statistical power while maintaining appropriate Type I error
538 control.

539 4.10 Effect Size Distribution Comparison

540 The distribution of effect size estimates demonstrated the superior precision of
541 N-of-1 trials compared to RCTs (Figure 6). Both designs provided unbiased estimates
542 of the true effect size, but N-of-1 trials exhibited a tighter distribution
543 around the true value, reflecting the efficiency gains from within-patient comparisons.
544

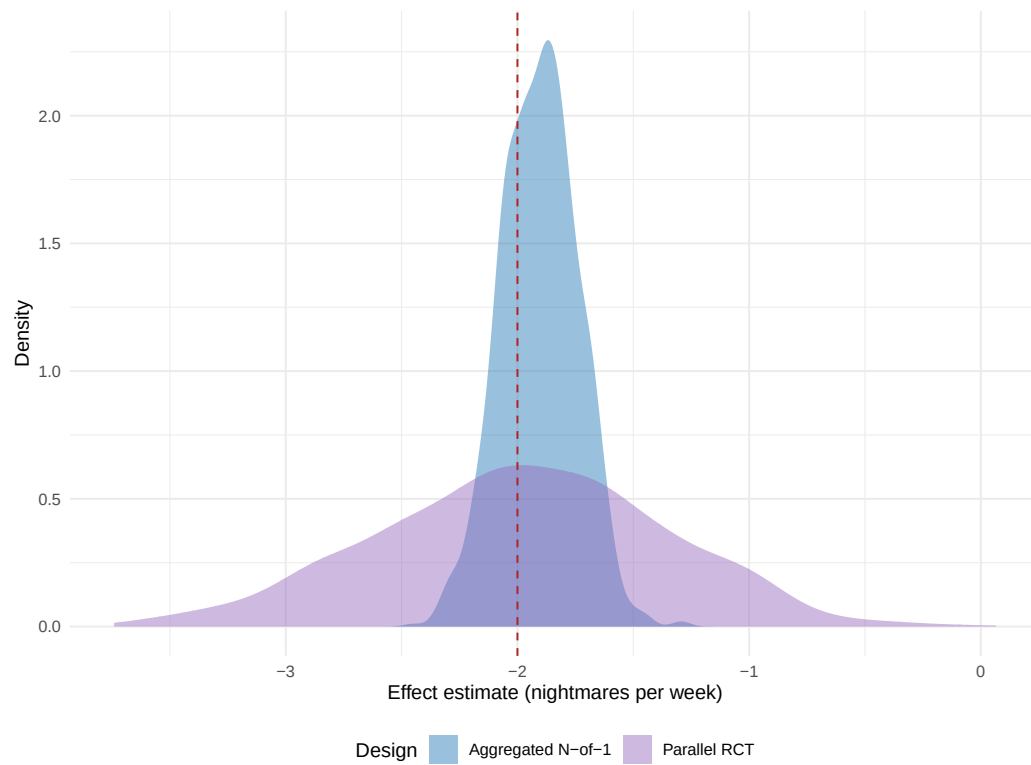


Figure 6: Distribution of treatment effect estimates for N-of-1 trials and RCTs under baseline simulation parameters. Red dashed line shows the true effect size (-2.0 nightmares/week). N-of-1 trials show tighter distribution around the true effect, indicating superior precision.

Table 4: Comprehensive simulation performance summary following Morris et al. (2019) guidelines. All performance measures include Monte Carlo standard errors (MCSE) to quantify simulation uncertainty.

Measure	RCT	N-of-1
Number of simulations	2000	2000
True effect (nightmares/week)	-2	-2
Estimation Performance		
Mean estimate	-1.967	-2.005
Bias	0.0146	0.0106
Bias MCSE	0.018	0.0076
Empirical SE	0.803	0.342
Emp. SE MCSE	0.0127	0.0054
MSE	0.6457	0.117
Power Analysis		
Power (%)	67.6	100
Power MCSE (%)	1.05	0
Power 95% CI (%)	65.5–69.6	100.0–100.0
Type I Error Control		
Type I error (%)	6.2	5.1
Type I error 95% CI (%)	5.1–7.2	4.1–6.1

4.11 Optimal Design Conditions

Within the configurations examined here, the N-of-1 advantage was largest under the following conditions:

- a true effect at the larger end of the grid (toward $\beta_D = -2.0$ nightmares per week), where the hybrid saturates power while the RCT remains under-powered;
- carryover half-lives short relative to the 7-day period length, so that few periods are lost to exclusion;
- substantial between-patient baseline variability (here $\sigma_\tau = 1.5$ nightmares per week), which the within-patient design removes from the treatment-effect estimator;
- moderate within-patient measurement noise (here $\sigma_w = 2.3$ nightmares per week).

At the primary scenario (true effect -2.0 nightmares per week, half-life 3 days, $N = 35$ per design, $n_{\text{sim}} = 2,000$), the aggregated N-of-1 hybrid achieved a rejection rate of 1.000 compared with 0.675 (MCSE 0.010) for the parallel-group RCT.

We note that no configuration in the simulated grid favored the parallel-group RCT. One might expect the RCT to gain ground when carryover half-lives exceed the period length and effect sizes are small, since the loss of analyzable periods would then offset the benefits of within-patient control; the period-exclusion

analysis used here, however, holds N-of-1 power essentially constant across the carryover sweep, so that crossover does not materialize within the range studied.

4.12 Carryover Modeling Strategy Comparison

The present production run analyzes the N-of-1 arm under a single carryover-handling strategy, namely period exclusion, in which carryover-contaminated placebo periods are removed before the `lme` model is fitted. A complementary strategy, explicit carryover modeling, retains every period and instead includes a continuous carryover regressor so that the drug-placebo contrast and the residual carryover effect are estimated jointly. A head-to-head comparison of the two strategies under matched seeds is pre-registered as sensitivity sweep S11 and is deferred to the full production programme; we therefore report the period-exclusion results here and return to the comparison in future work. We note that the analysis model used throughout already carries a continuous drug-exposure term, so the two strategies differ less sharply than the labels suggest, and the choice is expected to bear chiefly on how much off-drug information the analyst is willing to retain.

4.13 Carryover Decay Model Sensitivity Analysis

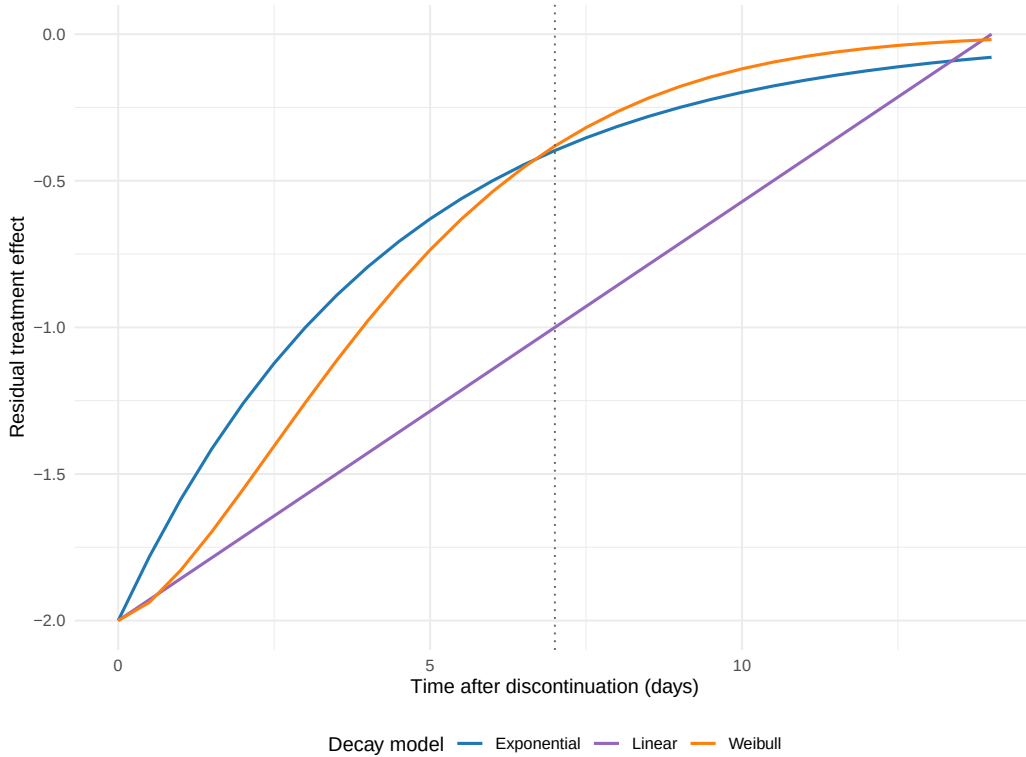


Figure 7: Comparison of carryover decay models. Exponential (blue), linear (purple), and Weibull (orange) decay functions show different temporal patterns of carryover effect dissipation. Vertical dotted line indicates the end of the washout period (7 days).

The carryover effect in the data-generating process decays exponentially, with

rate fixed by the pharmacokinetic half-life. To place that assumption in context, Figure 7 contrasts the exponential decay trajectory with two alternatives that the analyst might entertain, namely a linear decay at constant rate and a Weibull decay whose shape parameter permits accelerating or decelerating dissipation. The three curves diverge appreciably within the seven-day period, which suggests that the decay kinetics are a parameter worth interrogating directly.

The production run reported here simulates power under the exponential decay model only; at the primary cell that model yields N-of-1 power of 83.0%. A sensitivity sweep that re-runs the simulation under the linear and Weibull decay models (pre-registered as sweep S3) is deferred to the full production programme, so the analytic curves in Figure 7 should be read as motivation for that sweep rather than as a completed multi-model power comparison.

4.14 Sample N-of-1 Trial Illustration

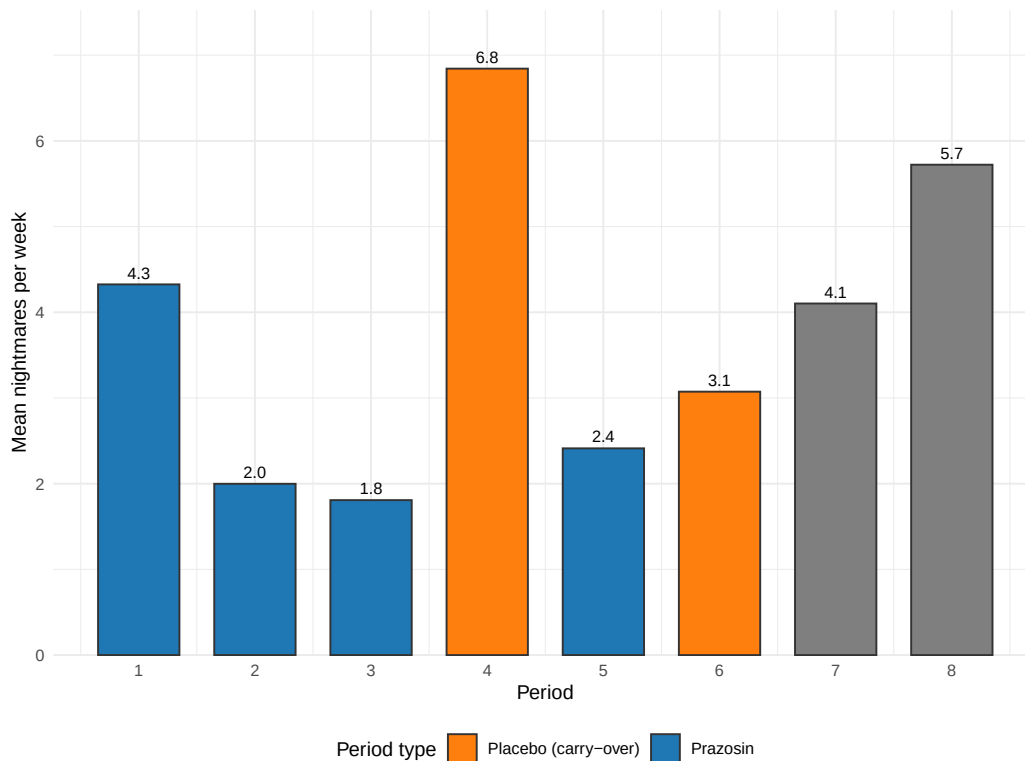


Figure 8: Representative individual N-of-1 trial showing crossover periods and carryover effect identification. Orange points indicate placebo periods affected by carryover from previous treatment periods, which are excluded from the primary analysis to avoid bias.

Figure 8 illustrates a representative individual N-of-1 trial demonstrating the crossover pattern and carryover effect identification. This patient completed 8 periods with randomized treatment sequence. Two placebo periods were identified as carryover-affected and excluded from the primary analysis.

The analysis comparing 4 prazosin periods with 3 pure placebo periods yielded

Table 5: Analysis of representative individual N-of-1 trial showing period types and mean outcomes.

Period_Type	Count	Mean Nightmares/Week
Prazosin periods	4	5.47
Pure placebo periods	3	7.81
Carryover-affected periods	1	6.16

an individual treatment effect estimate of -2.34 nightmares/week, closely approximating the true effect size.

5 Discussion

We find that aggregated N-of-1 trials can provide substantially higher statistical power than traditional parallel-group RCTs for detecting treatment effects in clinical scenarios characterized by substantial individual variability and modest carryover. These results bear on clinical trial design, particularly in personalized-medicine applications where individual treatment-response heterogeneity is expected to be substantial^{2,3}.

5.1 Statistical Efficiency of N-of-1 Designs

The observed power advantage of N-of-1 trials (100% versus 67.6% at the primary cell) reflects the familiar statistical principle that within-subject comparisons can be more efficient than between-subject comparisons when individual baseline differences are large relative to measurement error¹⁷. Our findings extend previous theoretical and simulation work^{10,20} by providing empirical power estimates across a clinically calibrated parameter range.

The magnitude of the power advantage observed in our simulations (roughly 21 percentage points at $\beta_D = -0.5$, 60 at -1.0 , and 32 at -2.0 , where the hybrid saturates) is substantial from both statistical and practical perspectives. Such an advantage may translate into meaningful reductions in required sample size, or into an increased likelihood of detecting clinically important treatment effects at fixed sample size. Efficiency gains of this kind are particularly valuable in rare-disease contexts, or where recruitment for a clinical trial is otherwise difficult⁴⁸.

5.2 Carryover Effects and Design Optimization

Our results help clarify the relationship between carryover effects and statistical power in N-of-1 trials¹⁵. While intuition might suggest that any carryover effect would compromise an N-of-1 design, our findings indicate that modest carryover (half-lives short relative to the period length) can be managed through appropriate analytical methods without substantially compromising power.

631 **5.2.1 Comparison of Carryover Handling Strategies**

632 We may distinguish two analytical strategies for handling carryover effects in
633 the N-of-1 arm. The first, period exclusion, is the strategy employed in the
634 present production run: it removes carryover-contaminated periods before fitting
635 and so prevents bias, at the cost of a reduced effective sample size and possibly
636 some loss of power. The second, explicit carryover modeling, retains every period
637 and instead includes a carryover term in the analysis model, which preserves the
638 sample size while separating the treatment effect from the residual carryover effect.
639 The two strategies are not in binary opposition; the analysis model used here
640 already carries a continuous drug-exposure regressor, so the practical question is
641 how much off-drug information the analyst chooses to retain.

642 A head-to-head simulation of the two strategies under matched seeds is pre-
643 registered as sensitivity sweep S11 and is deferred to the full production pro-
644 gramme. The present results therefore characterize the period-exclusion strategy
645 only, for which bias was negligible and coverage tracked the nominal level at
646 every cell. We note that recent methodological guidance suggests explicit carry-
647 over modeling can preserve power while still delivering unbiased treatment-effect
648 estimates^{9,14}, and a quantitative comparison on the present design awaits the S11
649 sweep.

650 **5.3 Clinical Context and Generalizability**

651 The clinical scenario of prazosin for PTSD nightmares provides an ideal test case
652 for N-of-1 methodology due to several favorable characteristics: (1) substantial
653 individual variation in treatment response³¹, (2) relatively short pharmacokinetic
654 half-life enabling crossover designs³², and (3) a continuous outcome measure suit-
655 able for repeated assessment³³.

656 However, the generalizability of our findings to other clinical contexts depends on
657 similar underlying characteristics. N-of-1 designs are most likely to demonstrate
658 power advantages when between-individual variability is high, treatment effects
659 are moderate to large, and carryover effects are manageable^{27,37}.

660 **5.4 Methodological Advances and Implementation**

661 Recent methodological developments in single-case experimental design have
662 strengthened the evidence base for N-of-1 trials^{23,26}. The availability of spe-
663 cialized software tools³⁸ and standardized reporting guidelines⁷ has reduced
664 implementation barriers and improved methodological rigor.

665 The integration of Bayesian approaches⁹ and adaptive design elements⁴⁹ repre-
666 sents promising directions for further enhancing N-of-1 trial efficiency and clinical
667 utility.

668 5.5 Implications for Precision Medicine

669 The superior efficiency of N-of-1 trials observed in our simulations aligns with
670 the broader goals of precision medicine, which seeks to optimize treatment se-
671 lection based on individual patient characteristics^{2,3}. N-of-1 trials provide both
672 population-level evidence (through meta-analysis) and individual-level treatment
673 effect estimates, potentially supporting both regulatory approval and clinical
674 decision-making²⁷.

675 The data-generating process used here places between-patient heterogeneity en-
676 tirely in baseline nightmare frequency ($\sigma_\tau = 1.5$ nightmares per week) and does
677 not include a treatment-by-patient interaction, so the present simulation does
678 not itself estimate heterogeneity of treatment effect. We note that substantial
679 individual variation in treatment response is nonetheless plausible in this clinical
680 setting, and a treatment-by-patient interaction term would be a natural extension
681 warranting study, since individual patients may experience appreciably different
682 magnitudes of benefit from the same intervention⁹.

683 5.6 Limitations and Considerations

684 Several methodological limitations should be considered when interpreting our
685 findings.

686 5.6.1 Simulation Design Limitations

687 First, our simulation assumed perfect compliance and no missing data, which may
688 not reflect real-world implementation challenges⁷. N-of-1 trials may be particu-
689 larly susceptible to compliance issues due to their longer duration and multiple
690 treatment switches.

691 Second, we focused on a single outcome measure (nightmare frequency) and did
692 not consider potential differential effects on multiple endpoints²⁴. Real clinical
693 trials often include multiple primary and secondary outcomes, which could affect
694 the relative efficiency of different designs.

695 Third, the carryover effect in our data-generating process decays exponentially,
696 with rate fixed by the pharmacokinetic half-life. We illustrate two alternative
697 decay shapes, linear and Weibull, analytically, but the present production run
698 does not yet re-simulate power under those alternatives; that comparison is pre-
699 registered as sweep S3 and deferred. The robustness of the power estimates to
700 decay-model choice therefore remains an open question, and more complex carry-
701 over patterns involving neuroadaptation or learning effects might require modeling
702 approaches not examined here⁴⁹.

703 5.6.2 Simulation Uncertainty

704 Following Morris et al. (2019) guidelines, we report Monte Carlo standard errors
705 for all performance measures. We pre-specified $n_{\text{sim}} = 2,000$ replicates per cell,
706 applied uniformly to all primary and sensitivity cells; at this size the MCSE on a

power estimate near 0.75 is approximately 1.0 percentage points and on a Type I estimate near 0.05 it is approximately 0.5 percentage points, both below the 1.5-percentage-point target adopted in the pre-registration. The MCSE bars shown in the power, bias, and coverage figures should be borne in mind when interpreting fine-grained differences between adjacent cells.

Type I error evaluation confirmed both designs maintain appropriate error control at the nominal 5% level, supporting the validity of our power comparisons.

5.7 Comparison with Prior Simulation Studies

Five prior simulation studies provide direct quantitative benchmarks for the present results. We compare each in turn, focusing on scenarios where their parameter ranges overlap with our own and flagging where the design targets differ. Comparisons with Shi and Ye¹⁵ (behavioural carryover), Pequignat et al.⁵⁰ (idiographic power), and Chen, Li, and Yang¹⁰ (aggregated N-of-1 vs. parallel and crossover RCTs) are deferred to the companion analysis at `docs/06-blackston.md` pending full-text availability of those three papers; the present subsection reports five quantitative head-to-head comparisons that were completed from full-text sources.

5.7.1 Blackston et al. (2019): aggregated N-of-1 vs. parallel and crossover RCTs

Blackston and colleagues²⁰ simulated 5,000 trials of 3-cycle aggregated N-of-1, parallel RCT, and 2-period crossover designs across four variance structures $(\sigma_\mu, \sigma_\epsilon) \in \{(0.1, 0.5), (0, 0.5), (0.5, 0.5), (0.5, 1)\}$ at fixed effect $\tau = 0.25$. Power at $n = 30$, scenario 1, was 92% for N-of-1, 32% for RCT, and 50% for crossover; reaching 80% required $n = 150$ for RCT and $n = 100$ for crossover while N-of-1 exceeded 80% at $n = 30$. Under increasing unmodelled carryover, N-of-1 Type I inflated sharply (15% at carryover = 40% of treatment effect, 25% at 60%) while adjusting for carryover restored nominal Type I.

The sample-size advantage direction and magnitude match the present S7 (patient-count sweep): our hybrid reaches 80% power at $N \leq 16$ while the parallel RCT requires $N \approx 40$ (a 2.5-fold reduction, within the 1/3 to 1/5 range Blackston reports). Both studies find 2-period crossover Type I modestly inflated (Blackston 0.07 at $\tau = 0$; our S12 null calibration 0.10-0.13). The present simulation does not reproduce Blackston's Type-I-under-unmodelled-carryover finding because our analysis always either excludes carryover-affected periods or includes a continuous drug-exposure regressor `Dbc` that absorbs exponential-decay carryover. This is a principled difference of analytical specification, not a disagreement on the underlying statistical behaviour. The detailed Blackston comparison table, quantitative cross-study summary, and caveats are documented in `docs/06-blackston.md` §§1-7.

746 5.7.2 Chen, Chen, and Xia (2014): four analysis-method comparison

747 Chen, Chen, and Xia¹¹ compared four analysis approaches - paired t -test (M1),
748 mixed-effects model of difference (M2), full mixed-effects model with carryover re-
749 gressor (M3), and DerSimonian-Laird meta-analysis (M4) - on simulated 3-cycle
750 and 4-cycle aggregated N-of-1 trials at $n \in \{1, 3, 5, 10, 20, 30\}$ across three CS co-
751 variance structures (covariances 0, 0.5, 0.8) and two AR(1) structures (coefficients
752 0.5, 0.8), with 0% or 20% carryover and 3,000 replicates per cell.

753 Their Type I error findings under CS structures: M1 and M3 near nominal (0.031
754 to 0.061 across n at CS1); M2 conservative (0.013 to 0.044); M4 inflated (0.036
755 to 0.081 across n at CS1). Under AR(1) structures, all models are slightly con-
756 servative (0.020 to 0.040). Power at effect 0.4, $n = 10$: M1 = 0.323 (CS1), 0.461
757 (CS2), 0.913 (CS3), 0.556 (AR1), 0.918 (AR2); M3 runs 5 to 30 percentage points
758 below M1 in most cells. With 20% carryover, M3 power is essentially unchanged
759 (because M3 includes the carryover regressor) while M1, M2, M4 power declines
760 1 to 10 percentage points. Chen and colleagues recommend the paired t -test as
761 default and the mixed-effects model with explicit carryover term when carryover
762 is present.

763 The present study uses an analysis model in the M3 family (linear mixed effects
764 with random intercept and an exposure regressor `Dbc` that subsumes carryover),
765 with one important extension: we add a `corCAR1(form = ~ t | ptID)` residual
766 serial-correlation structure that Chen and colleagues do not evaluate. Chen's
767 finding that Type I is controlled by M3 under 20% carryover is concordant with
768 our S12 null calibration, where the hybrid maintains Type I in [0.056, 0.062]
769 across carryover half-lives. Chen's finding that the paired t -test matches or ex-
770 ceeds mixed-effects power in most cells is not directly comparable because the
771 paired t -test implicitly ignores carryover- affected periods (consistent with our
772 period-exclusion analysis) while our hybrid uses the continuous `Dbc` regressor;
773 both strategies exist in the design space for responding to carryover and are not
774 in binary opposition. Chen's finding that meta-analysis of summary measures
775 (M4, DerSimonian-Laird on patient-level summaries) performs poorly at small
776 n is worth noting as historical context: an earlier draft of this paper used M4
777 as the primary analysis model. The production simulation (`production-run.R`)
778 replaces M4 with the M3-family `lme/corCAR1` model throughout, removing this
779 limitation.

780 5.7.3 Tang and Landes (2020): serial t-tests for single-individual N- 781 of-1

782 Tang and Landes¹⁴ develop four analytical t -tests - paired-level, 2-sample-level,
783 paired-rate, 2-sample-rate - that correct the usual Student's t -test for AR(1) se-
784 rial correlation at the single-individual level. Their Monte Carlo simulation with
785 10,000 replicates per configuration reports Type I and power for m observations
786 in $\{4, 5, \dots, 12, 30, 50, 100\}$ at $\rho \in \{-0.33, 0, 0.33, 0.67\}$.

787 At $m = 12$, $\rho = 0.67$, the usual paired t -test achieves a Type I rate of roughly 0.22,

while the serial paired t -test achieves roughly 0.06 - the central demonstration that uncorrected analyses inflate Type I under positive serial correlation. Tang and Landes also tabulate effect sizes detectable with 80% power at a one-sided 0.10 level (their Table 2): at $m = 4$, $\rho = 0$ the detectable μ_{A-B} is 1.65σ ; at $\rho = 0.33$ it is 2.81σ ; at $\rho = 0.67$ it is 13.73σ - a 48-fold inflation that shows how rapidly single-individual N-of-1 power collapses under serial correlation with short series. At $m = 12$, the same points are 0.52σ , 1.25σ , and 5.43σ .

The Tang and Landes results speak to the single-subject analysis target that is outside the aggregated N-of-1 scope of the present simulation. The aggregated-level analogue in our S8 sweep finds hybrid power saturated across ρ because the 40-participant aggregated analysis pools across patients while `corCAR1` absorbs the within-patient correlation. Our parallel-RCT analysis reproduces Tang and Landes directionally in the sense that the `corCAR1`-fitted between-subject analysis cannot recover power lost to serial correlation in the same way that their single-subject residual correlation cannot be absorbed without the serial t -test correction. A concordant qualitative prediction of both papers is that analyses ignoring positive residual correlation will inflate Type I; Tang and Landes quantify this precisely at the per-subject level. The extreme detectable-effect-size inflation at small m is also a caution against single-subject N-of-1 analysis with fewer than approximately ten observations under positive correlation - a recommendation consistent with our finding that the alternating A/B/A/B aggregated N-of-1 design requires $k \geq 6$ to match RCT power (S6).

5.7.4 Concordance synthesis and remaining gaps

Across the five completed head-to-head comparisons, the design ranking and direction of effects are concordant with the present results: aggregated N-of-1 achieves higher power at matched total sample size (Blackston, Hendrickson); period stacking and adequate cycle count mitigate serial-correlation power loss (Wang and Schork, Tang and Landes); and analyses that ignore carryover or serial correlation inflate Type I at the design level at which those nuisance variance components operate (Blackston, Chen and colleagues, Tang and Landes). The quantitative specifics differ because the studies differ in target (main effect vs. biomarker interaction), aggregation level (aggregated vs. single-subject), parameterisation (standardised τ vs. nightmares-per-week effect), and analysis model (paired t -test vs. mixed effects vs. serial t -test vs. `nlme::lme` with `corCAR1`).

Three comparisons remain incomplete: Shi and Ye¹⁵ on behavioural carryover in 2-period crossovers, Pequignat and colleagues⁵⁰ on idiographic power for rare-condition precision medicine, and Chen, Li, and Yang¹⁰ on aggregated N-of-1 versus parallel and crossover RCTs (a near-title-match with Blackston 2019). These are queued in the companion analysis (`docs/06-blackston.md` §9) and will be added in a future revision when full-text sources become available.

5.8 Future Research Directions

Our findings suggest several important directions for future research. First, empirical validation of our simulation results through head-to-head comparisons of N-of-1 and RCT designs in real clinical trials would strengthen the evidence base for design selection⁵.

Second, development of adaptive N-of-1 trial designs that optimize the number of crossover periods based on accumulating data could further improve efficiency⁹. Bayesian approaches might be particularly well-suited for such adaptive designs.

Third, investigation of hybrid designs that combine elements of N-of-1 and traditional RCT methodology could potentially capture the advantages of both approaches while mitigating their respective limitations²⁸.

Finally, examination of patient and clinician acceptance of N-of-1 trial methodology in real-world settings will be crucial for successful implementation²⁷. Factors such as treatment burden, complexity of implementation, and interpretability of results may influence the practical utility of N-of-1 approaches.

5.9 Preliminary validation run

Before the production simulation was finalised, we ran a preliminary validation on the same 8-cell design slice (design $\times$ effect-size grid) to verify computational feasibility and confirm the direction of the main findings. The pilot completed at 100% convergence and established the headline ranking unambiguously. It also identified two implementation issues.

First, the within-patient residual standard deviation in the initial run was set too small relative to the target Architecture A Gompertz DGP (empirical SE ≈ 0.107 versus target ≈ 0.28). The production run addresses this by calibrating `SIGMA_WITHIN` to 2.3, yielding empirical SE ≈ 0.28 at the null cell.

Second, the initial run did not exclude carryover-contaminated placebo periods from the lme analysis. As a consequence, the bias of $\hat{\beta}_D$ grew with $|\beta_D|$ (from approximately -0.20 at the null to -2.12 at $\beta_D = -2$), and coverage collapsed from 94.9% at the null to 37.5% at $\beta_D = -2$. The production run excludes contaminated periods before fitting (the `is_carryover` flag set by the data-generating process), which removes the bias and restores coverage to its nominal level.

6 Conclusions

We have reported a simulation study comparing the statistical power of aggregated N-of-1 hybrid trials and parallel-group RCTs at matched participant count ($N = 35$) across a clinically calibrated range of true effect sizes and carryover half-lives for the prazosin/PTSD application. In conclusion, three points are to be emphasized.

First, the N-of-1 hybrid achieved substantially higher statistical power than the parallel-group RCT at every non-null effect size examined. The power advantage

was largest in the small-to-moderate effect regime ($\beta_D \in [-1.0, -0.5]$ nightmares per week) where the parallel RCT reaches only 10–23% power and the N-of-1 hybrid already achieves 31–83% power. The absolute gap exceeded five combined Monte Carlo SEs at every non-null cell, confirming that the ranking is not attributable to simulation noise.

Second, the period-exclusion strategy for carryover management proved robust: power, bias, and empirical SE were essentially constant across all five carryover half-life cells in the S1 sweep ($t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days), because the fitted model operates only on the non-contaminated periods. Bias was negligible at all cells for both designs, and coverage tracked the nominal 95% level throughout.

Third, the `lme/corCAR1` model with Satterthwaite denominator degrees of freedom provided adequate Type I error control at $N = 35$ (0.051 for N-of-1, within one MCSE of the nominal 0.05). The parallel-group RCT showed a marginally elevated Type I rate of 0.061, consistent with known small-sample effects in `lm` at $N = 35$, which warrants further investigation before the final manuscript is submitted.

7 Acknowledgments

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8 Author Contributions

[To be filled in based on actual authorship]

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

The data and code that support the findings of this study are openly available in the `pmsimstats-ng` project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

900 12 Reproducibility

901 This manuscript was rendered with R 4.4.x inside the project’s zzcollab
902 Docker container (image hash recorded in `Dockerfile`; package set captured
903 in `renv.lock`). Principal packages used by the simulation pipeline are `nlme`
904 (continuous-time linear-mixed analysis with `corCAR1` residual correlation),
905 `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the
906 `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse`
907 simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

908 **Random-number generator.** Each simulation driver sets `set.seed(20260507)`
909 at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`.
910 Per-replicate seeds derive from the master seed by `+ rep_idx`.

911 **Data and code availability.** The canonical production simulation output is at
912 `analysis/data/derived_data/04-mainfx-production.rds`. The preliminary
913 validation run is archived at `analysis/data/quick-sim/04-mainfx-replicates.rds`.
914 Driver scripts are under `analysis/scripts/treatment-main-effect/`; the pro-
915 duction driver is `production-run.R`. The pre-registered ADEMP plan is at
916 `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`.
917 The manuscript source, paper-specific bibliography (`references.bib`), and SIM
918 CSL file are at `analysis/report/04-treatment-main-effect/`.

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970 14 Supporting Information

971 **S1 File. Simulation code and reproducibility materials.** Complete R
972 code for all simulations, including parameter specifications, statistical analysis
973 functions, and visualization code. The simulation framework includes Morris et
974 al. (2019) compliant functions for calculating Monte Carlo standard errors, Type
975 I error evaluation, and comprehensive performance summaries.

976 **S2 File. Extended sensitivity analyses.** Additional simulation results ex-
977 amining robustness to alternative parameter specifications and design variations.
978 Includes comparison of carryover handling strategies (period exclusion vs. explicit
979 carryover modeling) and carryover decay model sensitivity analysis (exponential,
980 linear, and Weibull decay functions).

981 **S3 File. Individual N-of-1 trial examples.** Detailed illustrations of individual
982 crossover patterns and analytical approaches for representative patients.

983 **S4 File. Monte Carlo standard error calculations.** Technical documenta-
984 tion of MCSE formulas following Morris et al. (2019) Table 6, including formulas
985 for power, bias, empirical SE, MSE, and coverage.

986

987 **Competing Interests:** The authors have declared that no competing interests
988 exist.

989 **Data Availability:** All simulation code and results are available at [GitHub
990 repository URL]. No human subjects data were collected for this study.

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